* Digital Communications *

SALEH RAMADAN

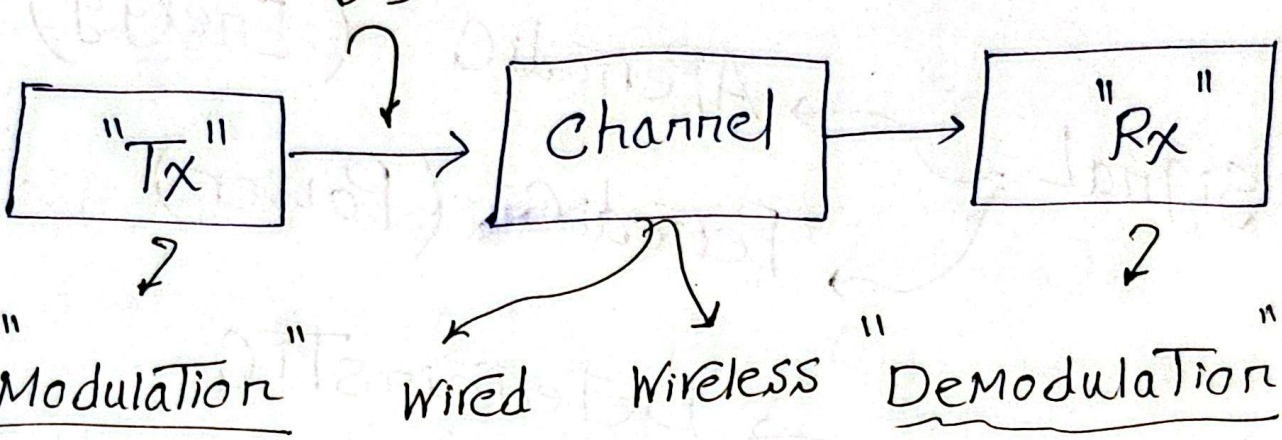
" Computer "

" Section 2 "

" وقل رب زدني علما "

* Introduction:

→ Communication System:



* Based on The Type of Signal, we will design Tx & Rx according to it

Signal

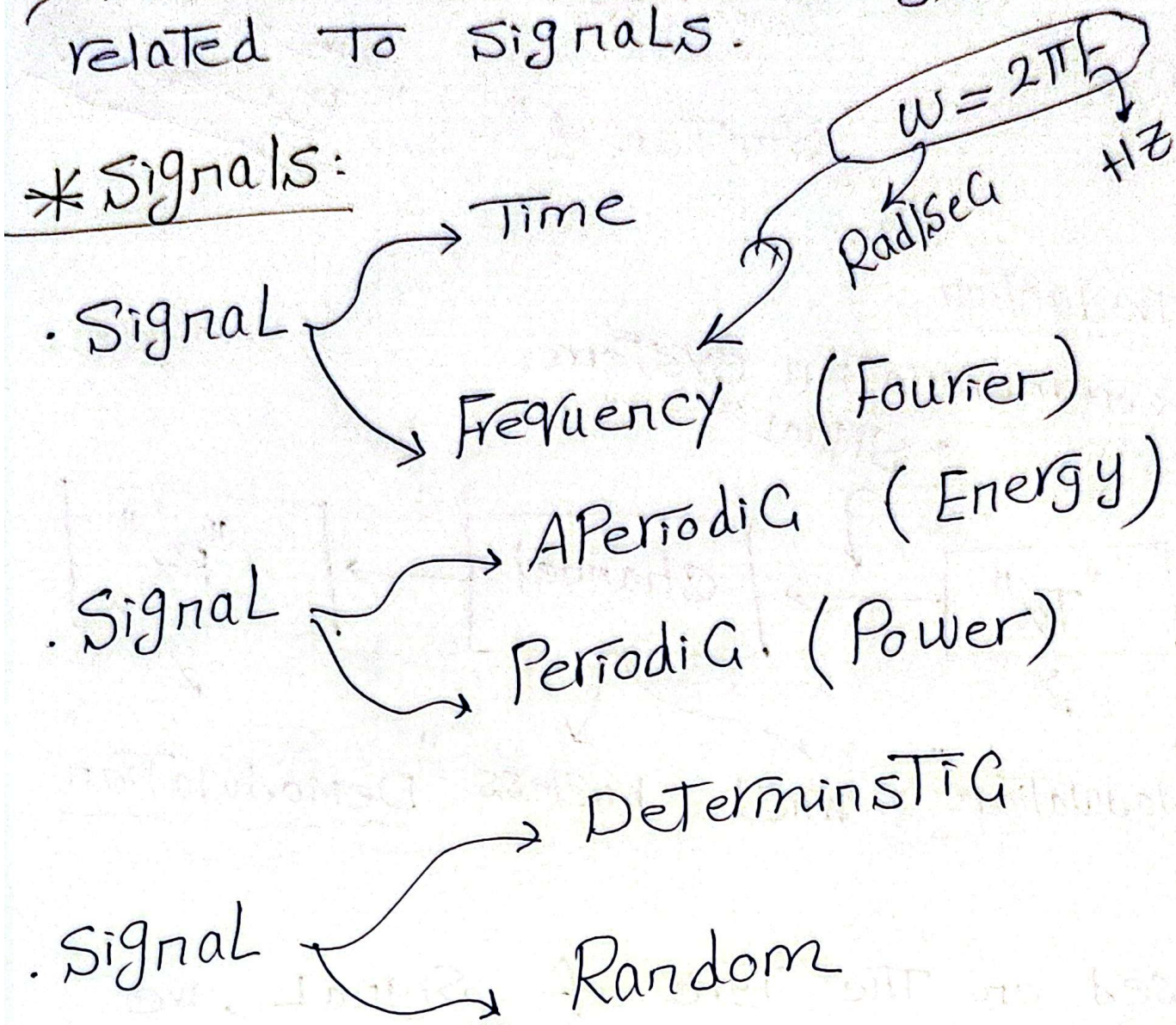
" Analog "
Continous Time
Continous Amplitude
(Analog Communication) (studied before)

" Digital Modulation " ↔ Communication

" Digital "
STream of Bits (01100)
(our study) this systems

→ Let's review some Things related to Signals.

* Signals:



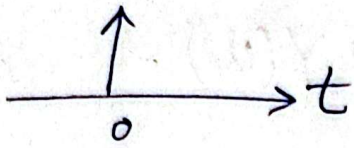
. BandwidTh (+ve Part in The SPecTrum)

B.W) = 2 B.W)
Pass Band — BaseBand

→ For Sinc Function: 1st NULL B.W

(3) · Delta Function:

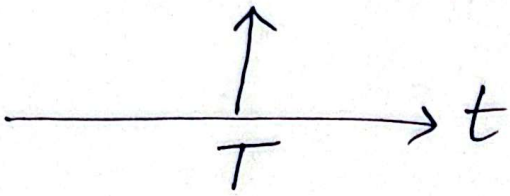
$\delta(t)$



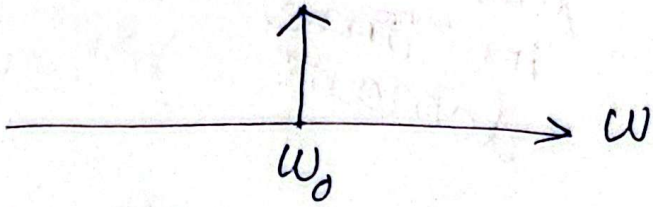
$\delta(\omega)$



$\delta(t-T)$



$\delta(\omega-\omega_0)$



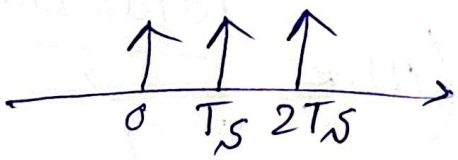
$\longrightarrow x(t)\,\delta(t-t_0) = x(t_0)\,\delta(t-t_0)$

$\longrightarrow x(t) \circledast \delta(t-t_0) = x(t-t_0)$

عند مكان Signal بـ Shift بعمل delta الـ

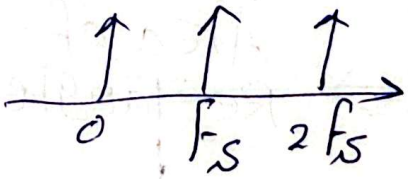
$\longrightarrow$ Impulse Train in Time domain

Impulse each T_s



F.T $\Longrightarrow$ Impulse Train in Freq. domain

Impulse each $f_s = \frac{1}{T_s}$



CamScanner

→ Fourier Transform: $x(f) \circledast Y(f)$

$$x(t)\,y(t) \xrightarrow{F.T} \frac{1}{2\pi}\left[X(w) \circledast Y(w)\right]$$

$$x(t) \circledast y(t) \xrightarrow{F.T} X(w)\,Y(w).$$

Multiplication in ~~Time~~ domain $\Longleftrightarrow$ Convolution in domain

- Famous Functions:

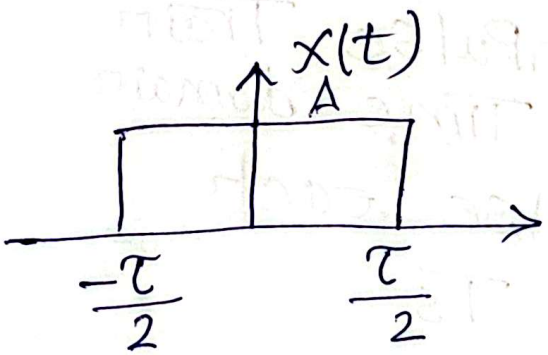
- $\delta(t) \longrightarrow 1$
- $\cos(w_o t) \longrightarrow \pi\left[\delta(w-w_o) + \delta(w+w_o)\right]$

$$\frac{1}{2}\left[\delta(f-f_o) + \delta(f+f_o)\right]$$

- $A\,\text{rect}\left(\frac{t}{\tau}\right)$

A: Amp. of Rect

τ: Total Base

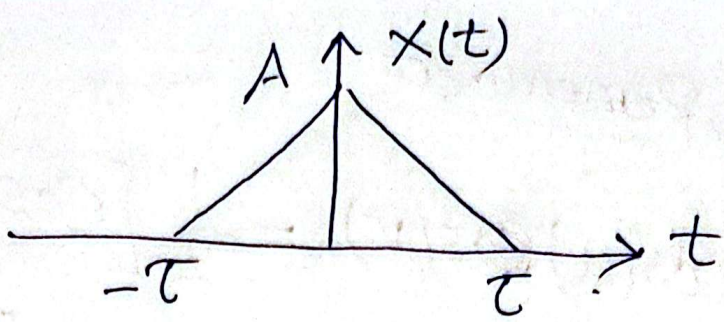


$$F.T \Longrightarrow \text{Area of Rectangle} * \text{SinC}\left(w * \frac{\text{Total Base}}{2}\right)$$

$$A*\tau * \text{SinC}\left(w * \frac{\tau}{2}\right)$$

- $\frac{A}{2}\,\Delta\left(\frac{t}{2\tau}\right)$

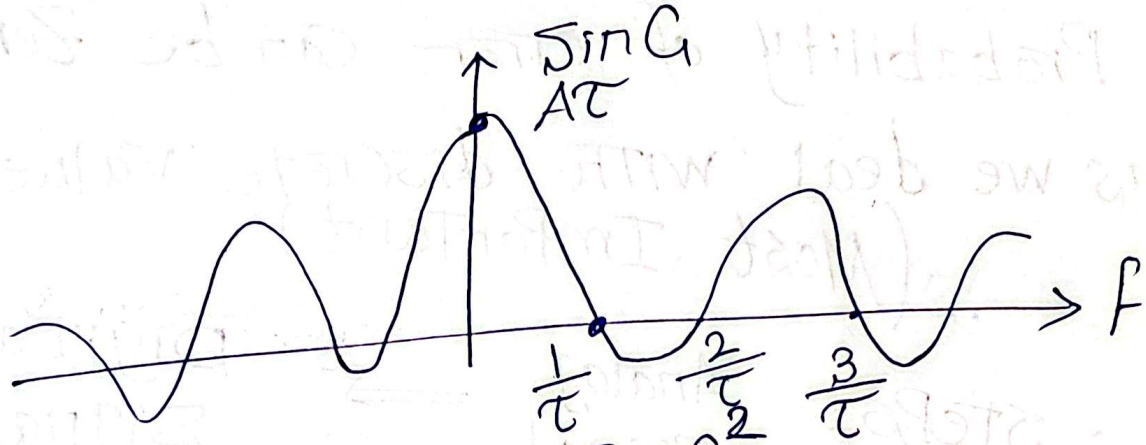
Amp of Traingle — Total Base

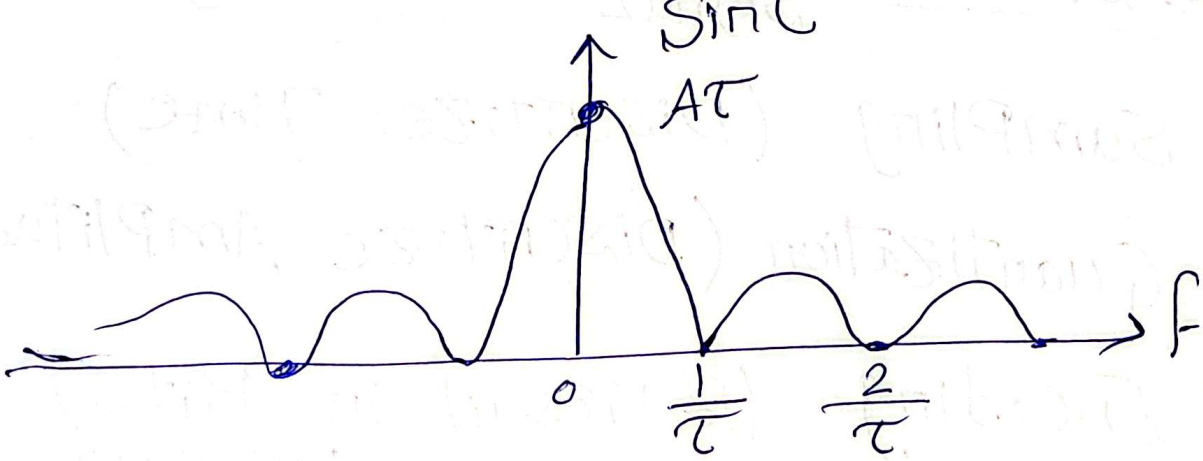


$$x(w) = \text{Area of Traingle} * Sinc^2\left(w * \frac{\text{Total Base}}{4}\right)$$

$$= A*\tau * Sinc^2\left(w*\frac{\tau}{2}\right)$$

- $Sinc(x) = \frac{\sin x}{x}$
- Zeros of Sinc or $Sinc^2$: $x = n\pi$





⑥

→ Remember:

$$\cdot \cos(u)\cos(v) = \frac{1}{2}[\cos(u-v) + \cos(u+v)]$$

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* Digitization:

→ Advantages:

① Inexpensive Circuit

② Mixing different types of data.

③ Security (Encryption)

⑤ Probability of error can be zero as we deal with discrete values. (Most Important)

→ Steps:

Analog signal ⟹ Digital signal

① Sampling (Discritize Time)

② Quantization (Discritize Amplitude)

③ Encoding (Represent in Bits) (01101101)

*Sampling:

Continous Time
Continous Amplitude

- Definition: Convert "Analog Signal" To "Samples" → Discrete Time, Continous Amplitude

- Parameters: T_s: Sampling Period "Sec"
  f_s: Sampling Rate (Frequency "Hz" "Samples/Sec"

$$\omega_s = 2\pi f_s \quad , \quad f_s = \frac{1}{T_s}$$

- Ideal Sampling:

$$x(t) \longrightarrow (\times) \longrightarrow x_p(t) = x(t)P(t)$$

$P(t)$ "Train of Impulses"

- Time: $P(t) = \sum_n \delta(t - nT_s)$

$$x_p(t) = x(t)P(t) = \sum_n x(nT_s)\,\delta(t - nT_s)$$

. Frequency:

$$\cdot P(f) = \frac{1}{T_s} \sum_{K} \delta(f - K f_s)$$

$$\cdot X_p(f) = X(f) \circledast P(f)$$

$$X_p(f) = \frac{1}{T_s} \sum X(f - K f_s)$$

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* Sketching:

. Time: ① draw signal in Time (State Main Points on x-AXis)

② Calculate "T_s"

③ Every T_s, Take Sample from The signal.

. Frequency: ① draw Spectrum of signal

② Calculate "f_s"

③ draw The Spectrum of signal each "f_s"

* Condition of Recover Signal:

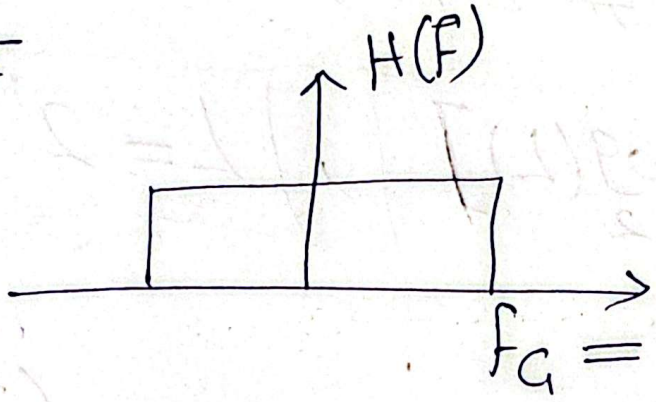
- T_S → small → Large Memory
- T_S → Large → Aliasing under Sampling (Lose Info from Message)

- Nyquist: $f_S \geq 2 * B.W)_{Message}$

↓

Nyquist Rate.

- use LPF



* Quantization:

- Definition: Convert signal From "Samples" → Continous AMP Discrete Time ⟹ "Levels" → Discrete AMP Discrete Time.

Parameters: L: # quantization Levels

Δ: Step size

$$\Delta = \frac{\text{Dynamic-Range}}{L} = \frac{A_{max} - A_{min}}{L}$$

If $A_{max} = |A_{min}| \Longrightarrow \boxed{\Delta = \frac{2A}{L}}$

***Encoding:** Represent Levels in Bits

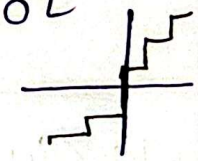
n: # bits / Sample

$$\boxed{n = \lceil \log_2(L) \rceil}, \quad \boxed{L = 2^n}$$

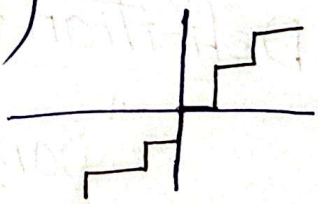
Ceiling

***Types:**

- Mid-Rise (Zero is not Level)



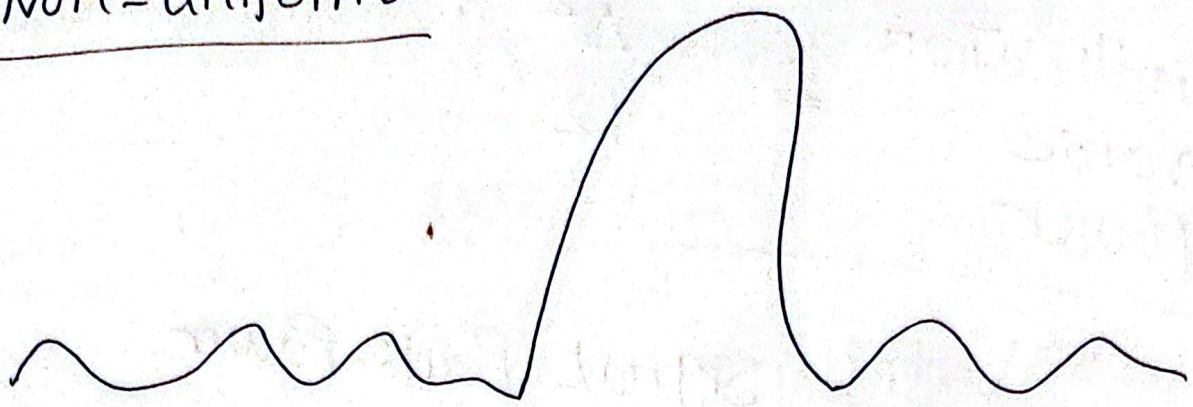
- Mid-Thread (Zero is a Level)



- Uniform (Equal Step Size)
- Non-uniform (Not Equal Step Size)

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→ Non-uniform:





* Performance:

$$q(t) = m(t) - m_q(t)$$

↙

quantization Error

- $q_{max} = \frac{\Delta}{2}$ ⇒

$$f(q) = \frac{1}{\Delta}$$

$$-\frac{\Delta}{2} \leq q \leq \frac{\Delta}{2}$$

$$q \sim U\left(-\frac{\Delta}{2}, \frac{\Delta}{2}\right)$$

CamScanner

$$\text{Quantization Noise Power} = \frac{\Delta^2}{12}$$

$$SNR|_{Peak} = \frac{\text{Signal Peak Power}}{\text{Quantization Noise Power}}$$

$$\text{Signal-Peak-Power} = A^2$$

$$SNR|_{Avg.} = \frac{\text{Signal_Average Power}}{\text{Quantization Noise Power}}$$

Signal Average Power
- Sinusoidal $\longrightarrow \frac{A^2}{2}$
- Random $\longrightarrow \frac{A^2}{3}$

→ For Non-uniform:

$$SNR \simeq \frac{3L^2}{[\ln(1+\mu)]^2}$$

*Note:
$SNR|_{dB} = 10\log(SNR|_{linear})$

*PCM (Pulse Code Modulation):

*Code carry Message:

$$R_b = \frac{1}{T_b} \quad \text{(bits/sec)}$$

R_b : Bit Rate

T_b : bit duration

$$T_s = n T_b \quad , \quad T_b = \frac{T_s}{n}$$

$$R_b = \frac{n}{T_s} \Longrightarrow \boxed{R_b = n f_s}$$

$$\boxed{B.W)_{PCM} \geq \frac{R_b}{2}}$$

*Note: For TDM: $\boxed{R_b = n f_x}$

f_x : #Signals * f_s